THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 3: **Metals and Non-metals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following will not react with dilute HCl? 1

(a) zinc **(b)** magnesium **(c)** iron **(d)** silver

2. Which of the following is correct about ionic compounds? 1

(a) They are soft **(b)** They have high melting points
(c) They are insoluble in water **(d)** They are gaseous

3. Iron articles are painted to prevent rusting because paint: 1

(a) reacts with iron **(b)** keeps air and water away
(c) makes iron reactive **(d)** makes iron heavy

4. A metal M reacts with cold water giving hydrogen and a base. M could be: 1

(a) copper **(b)** sodium **(c)** gold **(d)** silver

5. Electronic configuration of an element X is 2, 8, 2. It forms an ion of charge: 1

(a) -2 **(b)** +2 **(c)** 0 **(d)** +1

6. Zinc coating protects iron even if scratched because zinc: 1

(a) is less reactive than iron **(b)** is more reactive than iron
(c) is non-metallic **(d)** is heavy

7. Which of the following reactions will take place? 1

(a) $Cu + ZnSO_4 \longrightarrow CuSO_4 + Zn$ **(b)** $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$ **(c)** $Ag + CuSO_4 \longrightarrow Ag_2SO_4 + Cu$ **(d)** $Au + FeSO_4 \longrightarrow Au_2(SO_4)_3 + Fe$

8. An element with configuration 2, 8, 7 forms an ion: 1

(a) X^+ **(b)** X^- **(c)** X^{2+} **(d)** X^{2-}

9. The process of making oxide layer on aluminium thicker is: 1

(a) roasting **(b)** anodising **(c)** galvanisation **(d)** smelting

10. Which of the following is the correct order of reactivity? 1

(a) Zn > Mg > Cu > Fe **(b)** Mg > Zn > Fe > Cu
(c) Fe > Cu > Mg > Zn **(d)** Cu > Fe > Zn > Mg

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does calcium float while reacting with water? 2

12. Why does an ionic compound conduct electricity in aqueous solution but not in the solid state? 2

13. Why are ionic compounds usually soluble in water but insoluble in kerosene? 2

14. Why is it that copper does not react with dilute HCl but zinc does? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain with equations how aluminium reacts with oxygen and what protects it from further corrosion. 3

16. Explain the concentration of ore steps for a zinc sulphide ore. 3

17. Explain two reasons why iron is galvanised. 3

OR

Metal A reacts with dilute HCl but not B. A reacts with cold water but not C. Arrange A, B, C in order of reactivity and justify. 3

18. Why is it important to refine metals? Describe how impure copper is refined. 3

OR

Why do some metals like Na and K react vigorously with acid while Cu does not? Explain in terms of the series. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain, giving equations, the reaction of metals with — 5
(a) oxygen
(b) water
(c) acids using one metal for each.

OR

Explain how corrosion is prevented in — 5
(a) iron bridges
(b) aluminium utensils
(c) ships, with reasons.

20. Explain the method of extraction of — 5
(a) mercury
(b) zinc and
(c) sodium with reasons for the method.

OR

A student is given pieces of Fe, Zn and Cu and the solutions of $FeSO_4$, $ZnSO_4$ and $CuSO_4$. Predict the reactions and the reactivity order. — 5

END OF QUESTION PAPER — **Total: 40 Marks**

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